

The Effects of a Wetsuit on Biomechanical, Physiological, and Perceptual Variables in Experienced Triathletes

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Purpose: Wetsuits have been shown to change swim biomechanics and, thus, increase performance, but not all athletes are comfortable with their use because of possible modifications in motor coordination. The aim of this study was to evaluate the effects of wetsuit use on biomechanical, physiological, and perceptual variables. **Methods:** Eleven national- and international-level triathletes, familiar with wetsuit use, performed 7 × 200-m front crawl at constant preset speed twice, with and without a full wetsuit. The trunk incline (TI) and index of coordination (IdC) were measured stroke by stroke using video analysis. Stroke, breaths, and kick count, and timing (as breathing/kick action per arm-stroke cycle); stroke length (SL); and underwater length were analyzed using inertial-measurement-unit sensors. Heart rate (HR), rating of perceived exertion (RPE), and swimming comfort were monitored during the task. **Results:** A lower TI; IdC; number of strokes, kicks, and breaths; HR; and RPE for each 200 m were found in wetsuit compared with swimsuit condition. Higher values of SL and underwater length were found in wetsuit, whereas no differences were found in swimming comfort and timing of kicks and breaths. An increase for swimsuit condition in number of strokes and breaths, HR, and RPE was found during the task compared with the first 200 m. **Conclusion:** Wetsuit use reduces TI and, thus, drag; increases propelling proficiency; and shows lower fatigability, without modifying motor coordination, compared with swimsuit use at the same speed. The use of a wetsuit during training sessions is recommended to increase comfort and the positive effects on performance.

Keywords: open water, swimming kinematics, drag, fatigability, comfort

Swimming in open water leads athletes to face environmental challenges not typically present in the pool environment, such as swimming in packs, unpredictable waves and currents, salt water, absence of turns, and exposure to cold water temperatures.^{1,2} To prevent hypothermia, the use of a wetsuit is permitted in official competitions when the water temperature is below 20 °C in open water³ and in triathlons up to 1500 m below 22 °C in longer distances and below 24.6 °C in age-group competitions.⁴ A wetsuit is made of neoprene, a synthetic rubber that contains small bubbles of gas that have the primary role of reducing convective heat loss.⁵ The thickness of a wetsuit usually differs between the different parts of the athlete's body, but it cannot exceed 5 mm in triathlon,⁴ and it cannot be thinner than 3 mm in open-water swimming competitions.³ Other than protecting athletes from cold water exposure, wetsuit use has shown effects on buoyancy, friction drag,^{6–8} and propelling proficiency (expressed as stroke length [SL] and stroke index [SI]),^{7,9–14} leading to a performance enhancement.^{6–22} On the other hand, athletes have reported lower comfort

in the upper arms^{12,17} at the end of a swimming task and an inhibition in kicking action^{11,17,20} while swimming with a wetsuit.²² The amplitude of these effects seems to be affected by how familiar the athlete is with a wetsuit, by their swimming ability,^{7,12} and by the model of wetsuit used.^{12,17,18,22} Although wetsuit use has been reported to improve performance,^{6–22} its effects on motor coordination and comfort during a task simulating a competition effort are lacking.²²

The majority of the studies investigating the effects of swimming with a wetsuit have tested athletes only in short-distance^{7–9,12,16,17,19} or short-time^{13,15,18} tasks.²² However, both open-water swimmers and triathletes swim in open water from several minutes to hours (up to 25²³ and 3.8 km⁴ in official competitions, respectively). In such long efforts, the sensation of fatigue plays a fundamental role, and a change in biomechanical, physiological, and perceptual variables, defined as fatigability,²⁴ could affect performance. During swimming locomotion, the maintenance of correct motor coordination has a critical role in the optimization of the ratio between propulsion and energy expenditure.^{25,26} In this context, given the already reported effects on biomechanical,^{6–17} physiological,^{8,9,13,16,19,27} and perceptual^{12,13,17,18,27} variables, prolonged wetsuit use could have an additional role in fatigability and overall performance. Very few studies have tested athletes while swimming with a wetsuit for longer time tasks²² (20–75 min^{10,27–29}), reporting a mitigation in the decrease of core/skin temperature in cold water^{10,29} and a decrease²⁷ or no difference¹⁰ in heart rate (HR). Moreover, only one study evaluated the effects of fatigability on upper limb action, comparing the first and the last length during

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a preset distance of 1500 m.¹⁴ However, comparing wetsuit with swimsuit during a preset distance task could lead to artifactual results. Indeed, all the results found could be attributable to the higher swimming speed reported with a wetsuit^{7–9,12,14,16,17,22} instead of as a direct effect of the wetsuit per se.²²

The aim of this study was to evaluate the effects of a full-body wetsuit on biomechanical, physiological, and perceptual variables during a 7 × 200-m front-crawl training session at constant speed (equal to Olympic-distance race pace) in experienced triathletes. We hypothesized an effect of wetsuit use on biomechanical variables that would mitigate fatigability in front crawl at preset speed compared with swimsuit. Moreover, because of the tight fitting, we hypothesized that wetsuit use could decrease the comfort and motor coordination of triathletes.

Methods

Participants

Fifteen (5 females) national- and international-level triathletes (Tier 3–4³⁰) were recruited for the study. All triathletes were familiarized with the wetsuit use and with the rating of perceived exertion (RPE) scale (CR-10 modified, Italian version³¹). The CR-10 is routinely used during triathlon training camps³² and commonly utilized to monitor training load, and it is widely promoted by the federation during training courses for coaches. Four triathletes (1 female) were excluded (see “Methodology” section); therefore, only 11 triathletes were analyzed. Detailed information about the triathletes is provided in Table 1. Information regarding procedures was provided to each participant, and written informed consent was obtained. The study was approved by the Institutional Review Board of the University of Rome “Foro Italico” (reference: CAR 38/2020) and was conducted in accordance with the principles of the Declaration of Helsinki.

Design

Each triathlete performed the same swimming protocol twice, once with the wetsuit and once with the swimsuit, in a random and counterbalanced order (7 triathletes performed the wetsuit condition first and 8 the swimsuit) determined using computer-generated randomization (<https://www.graphpad.com/quickcalcs/randomize1/>). The sessions were performed just before the competitive season, when athletes mostly utilized wetsuits, at the same time of the day with at least 48 hours and no more than 7 days between sessions. The participants were instructed to maintain similar eating, sleeping, and training habits and avoid intensive exercise 48 hours prior to the tests.

Methodology

Before the first test session, subjects filled in an online survey (Google Form, Google) to collect individual information, such as age, height, training information, and wetsuit use habits (see Table 1). The tests were performed in an indoor pool (length = 33.33 m, water temperature = 28.4 ± 0.5 °C, and air temperature = 27.6 ± 0.8 °C) traditionally used by the triathletes. Prior to each test session, body mass and fat were estimated by an impedance balance (Mi Body composition Scale 2, Xiaomi).

A standard warm-up, self-paced swimming up to 15 minutes,^{9,33} was performed before each swimming test. The swimming test consisted of a 7 × 200-m front crawl constant-speed protocol, with 1-minute rest between repetitions, performed at the individual

average race speed of the Olympic distance triathlon (1500 m, see Table 1). A sound pacer (Tempo Trainer, Finis) was placed inside the swimming cap, and the swimmer followed the audio signal to synchronize with his preset speed. The test in the wetsuit condition was performed using the full model wetsuit (covering the whole body except for the face, hands, and feet) of each triathlete (details in Table 1). The subjects with a time difference between suit conditions in at least 1 repetition > 3% (~1.9 s) were discarded and not analyzed because of possible differences in biomechanical variables, as previously reported.³⁴

The test was recorded by an underwater camera (Hero4 Black, GoPro, 120 Hz) placed in the sagittal plane of the swimmer. At least 1 to 3 complete stroke cycles were recorded each time the triathlete passed in front of the camera. Black pen (swimsuit) or red tape (wetsuit) markers were applied on the following anatomical landmarks on both sides of the body: acromion process, interior angle of the scapula, great trochanter, fibula head, and lateral malleolus. Kinovea software (version 0.8.15, Joan Charmant & Contrib.) was used to manually analyze the video sequences frame by frame. Trunk incline (TI) was quantified using a video-based system.³⁵ The arm stroke phase events (entry, pull, push, and recovery) were identified using video analysis to estimate the stroke phase percentages and the index of coordination (IdC).³⁶ These variables were presented as mean value of the first and the seventh repetition of each test.

Three inertial measurement unit (IMU) sensors (WaveTrack Inertial System waterproof, Cometa, 128 Hz, accelerometer full scale: 16 g, gyroscope full scale: ± 2000 °/s) were placed on the occipital bone, on the right wrist, and 1 cm above the right lateral malleolus (Figure 1). The sensors were fixed with 2 swimming caps on the head and with biadhesive/coband tape on the limbs. The wrist IMU automatically recognized the instant that the wrist entered the water through the modulus of the signal output of the gyroscope (angular velocity). Due to the shock effect of the water drag on wrist–water touch, an artifact in the smooth gyroscope signal was used to detect wrist entry using the local maxima of the jerk. Furthermore, an algorithm computed the lateral face entry into and exit from the water surface by recognizing the peaks of the angular velocity of the occipital bone sensors in the mediolateral head roll. Finally, the downbeat end of the foot during the flutter kick was automatically detected by applying the method described in Fulton et al.³⁷ The following biomechanical variables were then calculated: breathing count (total, right and left side); percentage count of left/right breaths; timing of breaths (left and right) with respect to stroke cycle duration, starting with the right-hand entry; stroke count/lap; SL per lap; kick count/lap; timing of kicks (first, second, and third, when effectuated) with respect to stroke cycle duration, starting with the right-hand entry; underwater length from the push on the wall; index of synchronization³⁸; SI; and stroke–breath count ratio (ratio between stroke and breath count). The start and the end of the underwater phase were automatically recognized using the angular velocity of the ankle IMU for the wall touch and of the wrist IMU for first hand entry into the water, respectively. For more detailed information about the setup and analysis of IMU sensor data, we recommend referring to the article published by Fantozzi et al.³⁹

The HR was continuously recorded by a bend sensor for swimming (HRM-Tri and Forerunner 935, Garmin) during the test, and the mean value was estimated for each repetition. After each 200 m, the RPE was collected (CR-10 modified scale, Italian version³¹). The subjects also provided their swimming comfort during the rest phase by a scale ranging between –5 “Very bad,” 0

Table 1 Subject Characteristics, Training Information, and Wetsuit Details

ID	Sex	Age, y	Height, cm	Body mass, kg	Body fat, %	Perf.	Sp., m/s	Tri. P., y	Tri. T., h/wk	Swi. P., y	Swi. T., h/wk	Swi. T., km/wk	Swi. T., n/wk	Wetsuit exp., y	Wetsuit details (thickness range)
2	M	27	183	74.90	18.5	G.P.	1.39	11	21	11	8	23	4	10	Zoot–WikiWiki (2:5)
3	F	23	165	59.97	28.9	E.Jr.	1.28	11	24	11	9	25	6	8	Huub–Astana (3:3)
5	M	19	187	73.78	15.9	E.c. Jr.	1.38	10	24	11	8	26	6	5	Zoot–WikiWiki (2:5)
6	M	16	180	64.08	12.3	N.Ch.	1.14	6	25	11	5	15	3	2	Zoot–Force 1.0 (2:5)
7	M	17	173	78.48	23.9	N.Ch.	1.25	11	25	11	6	15	4	4	Huub–Aerious II (3:5)
8	M	17	174	66.23	15.5	N.Ch.	1.25	7	21	11	9	20	5	2	Tyr–Hurricane C3 (3:5)
9	F	24	162	49.73	21.8	W.Jr.	1.29	11	27	11	6	25	6	9	Yonda–Ghost (1.5:5)
10	F	20	173	61.65	27.8	E.c. Jr.	1.23	11	16	11	6	13	3	6	Huub–Acara (3:5)
12	M	17	175	75.25	20.3	N.Ch.	1.25	5	20	8	7	20	5	3	Zoot–WikiWiki (2:5)
13	M	18	182	73.60	17.8	E.Y.	1.41	8	23	11	9	26	6	3	Zoot–WikiWiki (2:5)
14	F	17	169	59.00	24.3	E.Y.	1.21	10	20	11	8	23	6	2	Huub–Aegis (3:5)
M		18.7 ± 3.8	179.4 ± 5.3	72.33 ± 5.20	17.7 ± 3.7		1.30 ± 0.10	8.3 ± 2.4	22.7 ± 2.1	10.6 ± 1.1	7.4 ± 1.5	20.7 ± 4.6	4.7 ± 1.1	4.1 ± 3.0	
F		21.0 ± 3.2	167.3 ± 4.8	57.59 ± 5.35	25.7 ± 3.3		1.25 ± 0.04	10.8 ± 0.5	21.8 ± 4.8	11.0 ± 0.0	7.3 ± 1.5	21.5 ± 5.7	5.3 ± 1.5	6.3 ± 3.1	
Total		19.6 ± 3.6	174.8 ± 7.7	67.0 ± 9.0	20.6 ± 5.3		1.28 ± 0.08	9.2 ± 2.3	22.4 ± 3.1	10.7 ± 0.9	7.4 ± 1.4	21.0 ± 4.8	4.9 ± 1.2	4.9 ± 2.9	(2.4:5) ± (0.6:0.6)

Abbreviations: M, Male; F, Female; Perf., best championship raced; G.P., ITU Grand Prix Senior; W.Jr., World Championships Junior; E.c. Jr., European Cup Junior; E.Y., European Championships Youth; N.Ch., Individual National Championships Senior; Sp., mean speed during Olympic-distance swimming fraction; Tri. P., triathlon practice; Tri. T., triathlon training volume; Swi. P., swimming practice; Swi. T., swimming training volume; Wetsuit Exp., wetsuit experience; n/wk, number of sessions per week. Note: the thickness range represent the higher and lower thickness value in mm independently to the body position reported by the individual company.

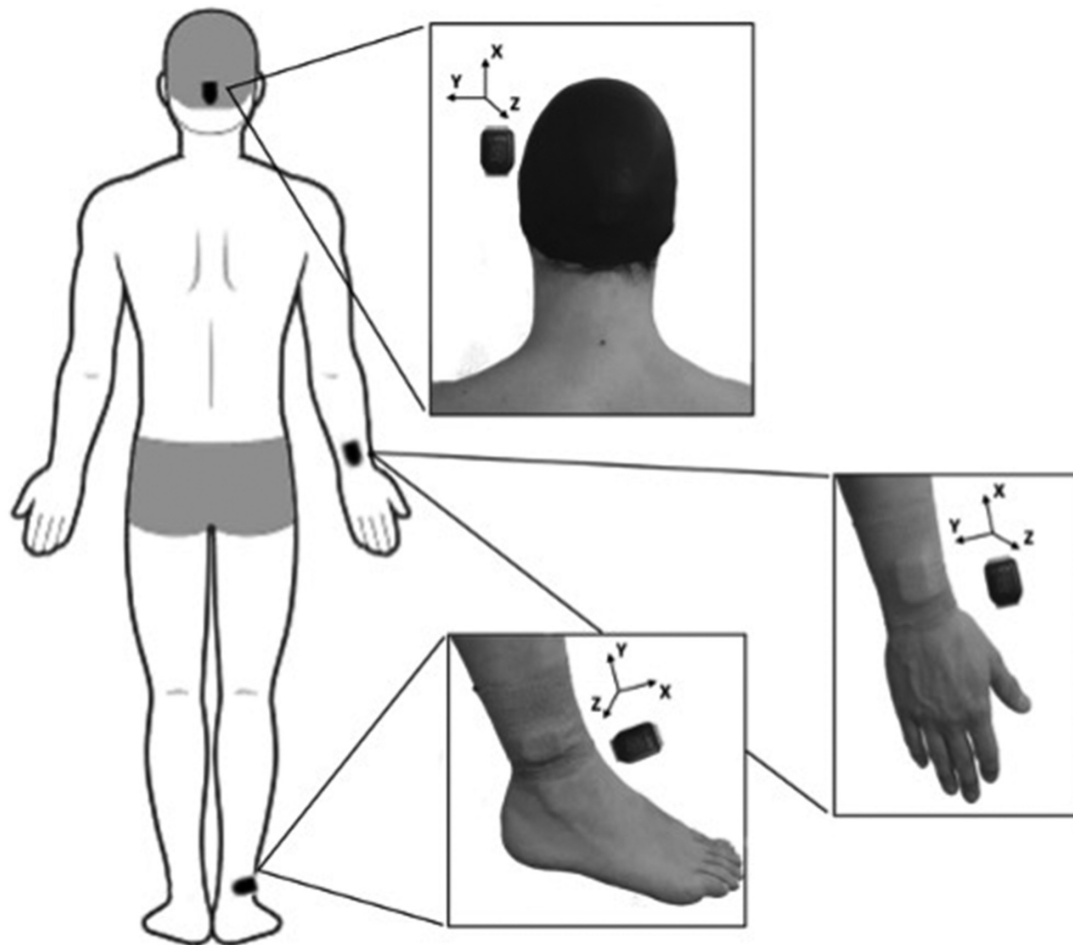


Figure 1 — Positioning of wearable inertial sensors located on the right ankle, right wrist, and head. The alignment of axes (x, y, and z) of the reference system in each sensor is also shown.

“Neutral,” and + 5 “Very good” (based on feeling scale⁴⁰), answering the question, “How do you feel your swimming comfort?”

Statistical Analysis

The sample size of 15 was estimated a priori using G*Power (version 3.1; Heinrich Heine Universität Düsseldorf), with a power of 0.7, alpha of .05, and *d* of 0.6 for *t*-test difference between 2 dependent means. The statistical package SPSS (version 25.0) for Windows OS was used for statistical analysis. To evaluate differences between suit conditions, a nonparametric Friedman test, both with Kendall W as effect size, was performed for all variables between conditions. A Wilcoxon test was performed post hoc, both with biserial correlation (*r*) as effect size, to assess pairwise differences between the conditions for the same repetition. To evaluate fatigability, a nonparametric Friedman test, both with Kendall W as effect size, was performed for all variables within each condition. A Wilcoxon test was performed post hoc, both with biserial correlation (*r*) as effect size, to assess pairwise differences between repetitions with the first one within the condition. The value of *r* was considered as: small (.100–.299), moderate (.300–.499), large (.500–.699), very large (.700–.899), and extremely large ($\geq .900$).⁴¹

The significance level was set at $P \leq .05$. Data are presented as median $\pm$ interquartile range.

Results

Triathletes completed the tests with a mean time difference of $1.09 \pm 0.47\%$ (~ 0.69 s) between suit conditions, with an average time of 158.02 ± 11.12 seconds and 158.72 ± 11.95 seconds for wetsuit and swimsuit condition, respectively. Four of the recruited triathletes achieved a time difference $>3\%$ ($5.0 \pm 2.0\%$, corresponding to 7.73 ± 3.07 s) between conditions, with the higher times performed in the swimsuit condition. These 4 subjects were discarded and not analyzed, as previously stated.

Suit Condition

Lower TI ($P < .001$; Figure 2) and a longer underwater phase (wetsuit 3.08 ± 0.46 m, swimsuit 2.71 ± 0.50 m; $P = .004$) were found in wetsuit condition. Number of both strokes ($P < .001$) and kicks ($P < .001$) was lower in wetsuit, with an associated higher SL ($P < .001$) and SI ($P < .001$) compared with swimsuit (Figure 3), and no difference in timing of kicking was found. Although a catch-up coordination was observed in both conditions, lower values of IdC were found in wetsuit compared with swimsuit ($P < .001$), both with a higher percentage in the no propulsive phases ($P < .001$), entry + catch (A: wetsuit $32.35 \pm 8.87\%$, swimsuit $32.10 \pm 7.64\%$) and recovery (D: wetsuit $28.63 \pm 8.72\%$, swimsuit $25.77 \pm 8.07\%$), and a lower

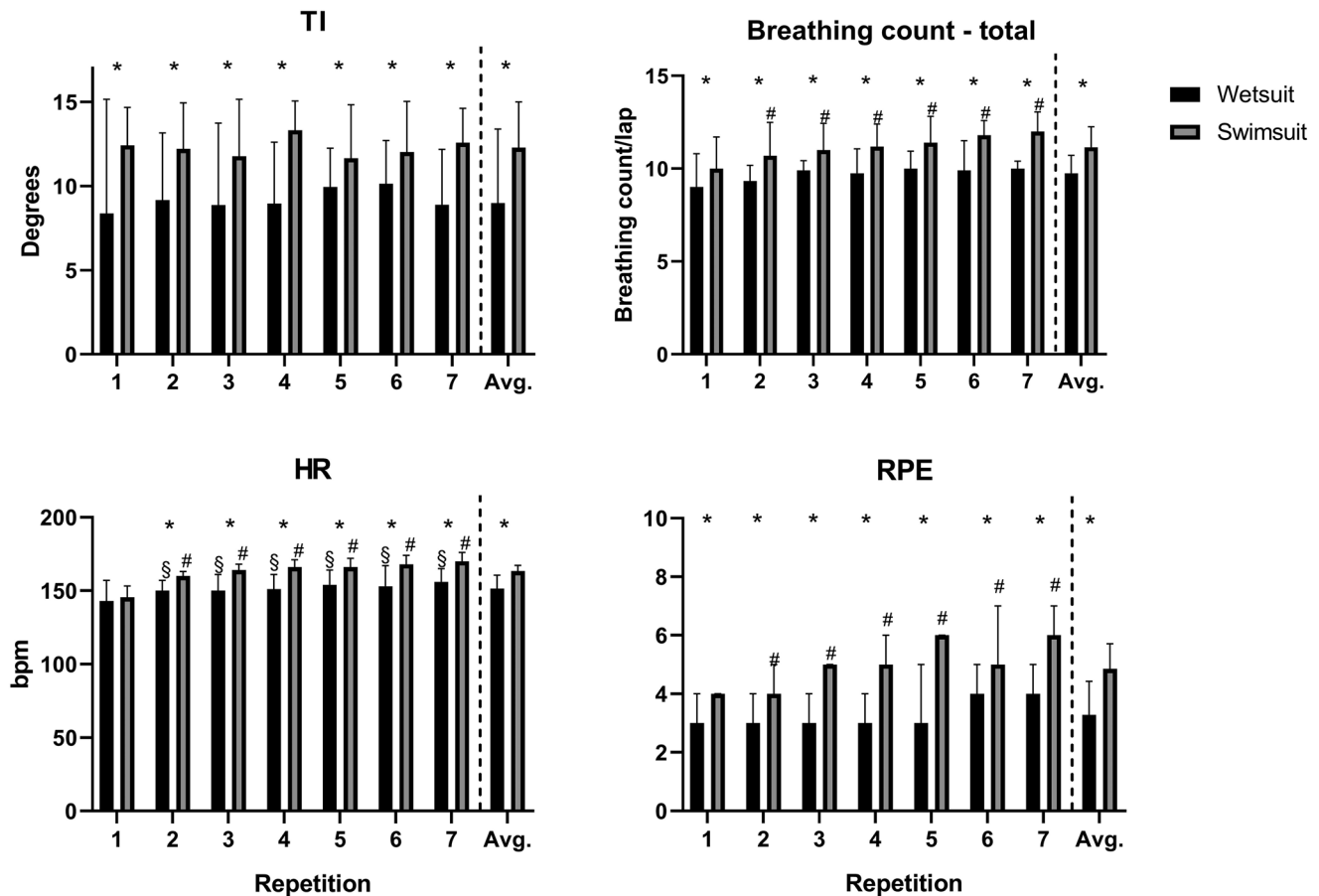


Figure 2 — TI, breath count per lap, HR, and RPE for each repetition and Avg. in wetsuit and swimsuit. Avg. indicates average; bpm, beats per minute; HR, heart rate; RPE, rating of perceived exertion; TI, trunk incline. *Significantly different between conditions. §Significantly different from repetition 1 in wetsuit. #Significantly different from repetition 1 in swimsuit.

percentage in the propulsive phases ($P < .001$), pull (B: wetsuit $16.98 \pm 2.25\%$, swimsuit $19.22 \pm 3.00\%$) and push (C: wetsuit $23.35 \pm 1.57\%$, swimsuit $24.65 \pm 4.17\%$) (Figure 3). No differences were found in index of synchronization (wetsuit 0.06 ± 0.22 , swimsuit 0.01 ± 0.12 ; $P = .911$). The total (Figure 2) and right side number of breaths were lower in wetsuit (both $P < .001$) but not on the left side ($P = .726$). The higher stroke-breathing count ratio found (wetsuit 1.38 ± 0.43 , swimsuit 1.28 ± 0.32 ; $P < .001$) confirmed the difference in number of breaths between conditions. However, no difference was found in the total percentage of right (wetsuit $62.9 \pm 24.09\%$, swimsuit $70.03 \pm 26.35\%$) and left side (wetsuit $37.85 \pm 23.63\%$, swimsuit $31.14 \pm 26.13\%$) breathing or in the timing of breathing ($P = .105$). Higher HR and RPE were recorded in swimsuit (both $P < .001$) with no differences in swimming comfort (wetsuit 0.1 ± 1.9 , swimsuit 0.7 ± 2.0 ; $P = .206$) (Figure 2).

Fatigability

A fatigability effect was found for HR in both swimsuit and wetsuit (both $P < .001$) and for count of strokes and breaths (both $P < .001$), SL ($P < .001$), stroke-breath count ratio ($P < .001$), and RPE ($P = .001$) in swimsuit condition only (Figures 2 and 3).

Detailed data and statistical analysis results are provided in [Supplementary Material](#) (available online).

Discussion

Wetsuits are permitted in open-water and triathlon events, depending on water temperature, age group, and distance to be covered, with the main purpose of preventing hypothermia.²² Previous research has shown an improvement in performance with wetsuit use by increasing buoyancy and gliding length and by decreasing energy cost.^{8,9,13,18,19} However, athletes have reported higher discomfort while swimming with a wetsuit, and some technical changes have been reported, such as an inhibition in kicking action.^{17,20,21} Our results have further strengthened what is already known by highlighting that wetsuit use decreases TI and drag, leading to a delayed increase of effort. Contrary to our hypothesis, motor coordination (ie, stroke coordination model, upper- and lower-limb coordination, and breath timing in a stroke cycle) was not affected by the wetsuit and neither was swimming comfort in experienced triathletes.

During a swimming activity, the different density between legs and chest leads to an inclination of the body alignment on the sagittal plane of the swimmer, also known as “sinking legs,” that increases resistance and reduces swimming speed.⁴² The results of our study highlight the effect of wetsuit use on reducing TI (~28%, $>3^\circ$) and, thus, the frontal area presented by the swimmer to the water that has an effect on the form component of active drag.⁶ Toussaint et al,⁶ in their pioneer work in 1989, showed a 14%

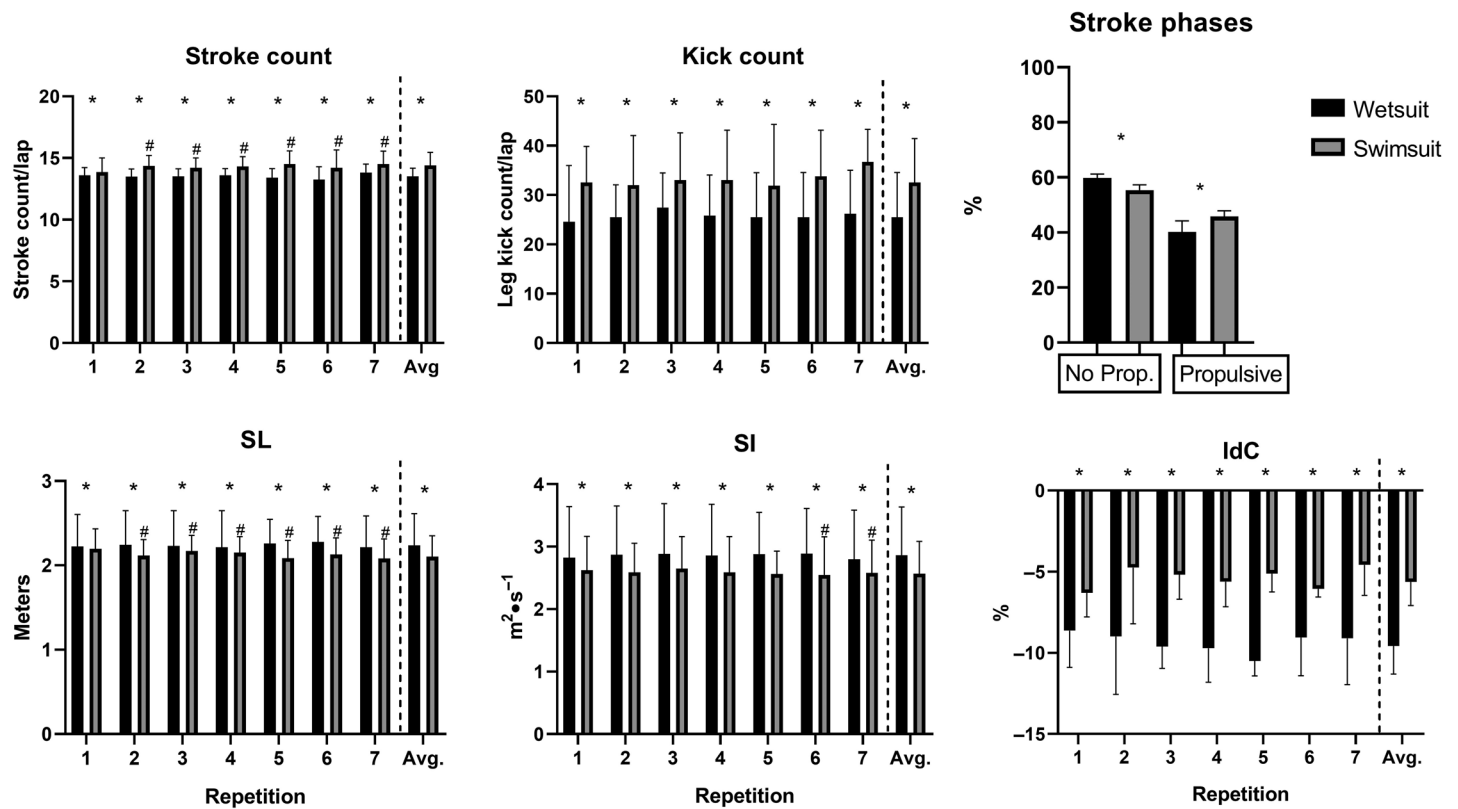


Figure 3 — Stroke (right side) count per lap, leg kick (right side) count per lap, stroke phase percentages (no propulsive = entry + catch and recovery, propulsive = pull and push), SL, SI, and IdC for each repetition and Avg. in wetsuit and swimsuit. Avg. indicates average; IdC, index of coordination; No prop., no propulsive; SI, stroke index; SL, stroke length. *Significantly different between conditions.

reduction in active drag (estimated with the measurement active drag system) while swimming in wetsuit condition at 1.25 m/s. Other studies, instead, showed no differences due to wetsuit use in active drag (measured by the perturbation method) when evaluated during a maximal sprint.^{8,19} However, the speed in the wetsuit condition during a maximal sprint was higher (~5%⁸) compared with the swimsuit condition, and this could justify the fact that no differences were found in active drag.¹³ Moreover, we can hypothesize that friction drag is also decreased in wetsuit, as suggested by the longer underwater phase found in the current study (~13.7%), attributable to the water-repellent properties of the wetsuit surface. Similarly, friction drag has been shown to decrease with wetsuit use, inversely to the speed, when athletes were towed in a prone position.⁷ As a consequence, there seems to be a direct effect of wetsuit use in decreasing drag during swimming, at least in the form and friction components.

Due to the decreased drag in wetsuit condition, the propelling proficiency seems to increase by reducing stroke rate and increasing SL and SI at the same swimming speed. Our results support previous findings wherein open-water swimmers and triathletes swimming with a wetsuit showed technical adaptations that reduced stroke rate and increased SL.⁹ Although the IdC was classifiable as a catch-up model in both conditions, our results showed lower percentage values in wetsuit condition compared with swimsuit. The lower propulsive phases indicate a shorter period of propulsive force application, thus low power generation at the same swimming speed using a wetsuit.³⁶ However, other studies involving triathletes or open-water swimmers showed different results^{11,13,16} depending on speed, duration, and if

swimming in a flume or in a swimming pool. It has to be pointed out that most of the studies in the literature did not test subjects over a preset speed, leading to artifactual results attributable to the higher swimming speed reached in wetsuit condition.^{9,16,22}

Regarding the effects of wetsuit on number of kicks, our results confirm the perceived decrease in kicking frequency previously reported by athletes using a wetsuit^{20,21} and in accordance with what has already been reported during an 800 m¹⁷ task but not in shorter tasks.¹¹ Leg kicks have the main effect of reducing TI during crawl swimming⁴³; the decrease of TI by wetsuit use brings about a consequent reduction in number of kicks needed to maintain the same position in the water. Moreover, the reduction of kicking frequency is profitable in both triathlon and open-water races, which require cycling and running after the swimming fraction in triathlon and very intensive finishing strategies in long open-water races.^{2,22}

Motor coordination, in particular the IdC model, index of synchronization value, timing of kicking, and breathing appeared not to be affected by wetsuit use, indicating that aspects of motor control acting on technique are not related to the suit used. An unchanged effect in kick adjustment by wetsuit use is favorable to athletes during swimming in adverse environmental conditions where waves might change the arm-leg coordination related to the effects of the moving body of water.²⁶ Moreover, a correct synchronization between breathing and stroke has been shown to be crucial to apply a great impulse of force, and its correctness is related to the relative swimming speed and performance level.²⁵

Enhancing buoyancy and decreasing drag seem to lead to a decrease of both physiological variables (lower HR and number of

breaths) and RPEs (lower RPE) at the same swimming speed while using a wetsuit compared with swimsuit. RPE increase throughout exercise is recognized as a major feature of the sensation of fatigue as well as playing a pivotal role in regulating pacing during exercise.⁴⁴ Moreover, it has been reported that respiratory frequency is the best correlate of RPE during self-paced maximal effort exercise irrespective of the intermittent or continuous nature of the protocol.⁴⁵ During a swimming activity, in particular during front crawl, the breathing action disrupts the body alignment, increasing drag and reducing SL.²⁵ Interestingly, a lower breathing frequency was found in more skilled swimmers compared with less skilled.²⁵ The current study is the first to investigate the effects of wetsuit on breathing, showing a lower number of breaths compared with swimsuit. The differences showed in RPE and HR are similar to what was previously shown at a speed of 1.31 m/s but not at lower speeds^{13,18} or at preset distance^{9,12,17} or time.¹⁰ However, a preset distance or time test implies a difference in speed between swimsuit and wetsuit^{9,10,12,17}; therefore, lower values of RPE or HR may be expected when normalizing by speed. Moreover, the preset speed tests were performed in a flume for 5 minutes for each trial,^{13,18} which may not be long enough to detect changes in RPE and HR. Indeed, we found no differences in HR or RPE during the first 200 m, but thereafter, both increased in the swimsuit condition but not in the wetsuit.

The choice of our test was to evaluate not only differences between swimsuit and wetsuit on overall performance but also how the use of wetsuit could mitigate the fatigability throughout a longer task. In detail, in the swimsuit condition, more variables (ie, RPE, HR, number of strokes, and breaths) increased from the beginning to the end of the 7 × 200-m, whereas in the wetsuit condition, only HR increased during the trial with no difference in RPE and number of strokes and breaths. We found no changes in stroke phase percentage during the task, whereas an increase in the propulsive phase was found between the first and the last length of 1500 m in wetsuit but not in swimsuit.¹⁴ However, the task was conducted at maximal speed, and it is unlikely that the swimming speed both between conditions and between the first and last length were the same. Recently, Rois et al²⁷ reported a decreased HR in wetsuit condition compared with swimsuit while swimming in a 25-°C flume for 75 minutes with no difference in RPE. However, the authors suggested that the increase in HR during the swimsuit condition was mainly due to compensation for heat generation rather than an index of intensity. Athletes were, in fact, swimming at an intensity of 70% of their critical velocity with a final RPE of 3 that corresponded to a subventilatory threshold 1 intensity.⁴⁶ Our triathletes, instead, reported an average RPE of 5, indicating a swimming intensity between the 2 thresholds.⁴⁶ The higher intensities reported in the current study might explain the variation in technical variables (ie, number of strokes and SL) and the increase of physiological (ie, HR and number of breaths) and perceptual (ie, RPE) variables in swimsuit, contrary to the steady state shown in the wetsuit. The fatigability on biomechanical variables (ie, decreasing SL, increasing stroke rate, increasing kicking speed, increase of IdC value but not changing the model) has been reported during a classic 7 × 200-m incremental protocol between each repetition.³⁴ In particular, these variations were evident at speeds above the lactate inflection point where a technique reorganization was shown by athletes to overcome increased hydrodynamic drag. This confirms the higher relative intensity sustained by our triathletes while swimming with a swimsuit compared with a wetsuit at the same swimming speed (Figure 3).

The current study was the first to evaluate swimming comfort in wetsuit and swimsuit during swimming activity.²² Contrary to expectations, no difference in comfort was found. Based on our preliminary survey, triathletes reported disliking wetsuit use mainly for upper arm discomfort during swimming (82%) and that they were forced in its use by rule restrictions (82%) and by the fact that opponents would use it (100%). However, there was no difference in swimming comfort with or without wetsuit, probably due to their familiarization with wetsuit use, as Table 1 shows. In fact, previous studies showed that pool swimmers reported a lower comfort because they were more unfamiliar with wetsuit use compared with triathletes.¹² It is important to note that there is no clear indication in the literature on the definition of familiarity of an athlete with a wetsuit.²² Our study is the first to report the numbers of years each athlete has been training/competing with a wetsuit. In particular, the triathletes of the current study have been regularly using wetsuit (from 2 to 10 y) in a pool or open water in addition to competitions. The current study was carried out just before the competitive season, when athletes mostly utilize wetsuits in training and competition.

Practical Applications

Swimming with a full-body wetsuit during a 7 × 200-m front crawl at preset speed, corresponding to the swimming race pace of an Olympic distance triathlon, leads to a reduction of TI and drag, at least in the form and friction components, and, thus, to a delay in fatigability. Interestingly, no changes in motor coordination were found, which confirms that triathletes may utilize wetsuit also during training sessions. Indeed, triathletes with experience in wetsuit use did not perceive high discomfort and benefited in terms of performance and reduced fatigability by wetsuit use.^{7,12,22} Therefore, the use of wetsuit is recommended also in training sessions to increase familiarization, without concerns about a possible negative effect on coordination. However, the amplitude of the effects might differ based on athletes (ie, open-water swimmers and triathletes, elite and recreational level, experience in wetsuit use), and more studies are necessary to highlight possible differences.

Our study has some limitations. The study was conducted in an indoor swimming pool at water temperature of 28.4 ± 0.5 °C due to the unfeasibility of cooling the water. In contrast, the athletes usually compete in an open-water environment where wetsuit use is allowed only at temperatures below 20 to 24.6 °C.^{3,4} Indeed, recent studies report differences in physiological variables when comparing swimming in cold temperature^{15,27} with a typical pool environment and also in biomechanical variables compared with open water.⁴⁷ However, it seems that different water temperatures (at least between 26 and 18 °C) do not affect biomechanical parameters.¹⁵ Our aim was to evaluate the presence of fatigability in biomechanical, physiological, and perceptual variables during a protocol similar to a training situation. Studies conducted in an open-water environment, simulating competition settings, are, thus, recommended.²² We recruited only 15 young triathletes, but they were of international level and with experience in wetsuit use. However, 4 triathletes were excluded because they were not able to replicate the speed maintained during the wetsuit session, and analysis was performed on only 11 subjects. The training session protocol of 7 × 200-m front crawl constant-speed interval with a 1-minute rest was designed to allow triathletes to correctly perform the task and to be able to compare the variables. Second, it allows for in-between measurements (ie, RPE and comfort). Our

main intent was to compare at the same speed the selected variables but also to measure the effect of fatigue, which we could obtain only by increasing the number of repetitions. Further studies performing a long continuous task are recommended, as already recommended in a recent review on the topic,²² to confirm our results obtained during this simulated training session. Finally, the stroke and kick data were estimated by IMU sensors located on the right limbs only. In following studies, we suggest locating IMU sensors on both sides for refining the estimation of technical variables.

Conclusions

Using a wetsuit during a swimming task simulating a competition effort reduces the TI and, thus, drag, compared with a swimsuit at the same speed. As a consequence, triathletes show an increased SI and SL, both indexes of propelling proficiency and a lower fatigability, without modifying motor coordination. The use of a wetsuit during training sessions is also recommended to increase comfort and the positive effects on performance.

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